

CSNS RCS IPM — Ion-mode scan results

Code: Virtual-IPM 2.3.1. **Reference:** NIMA **1092** (2026) 171809 (CSNS RCS IPM). **Campaign:** round-beam ion-mode matrix, **2070 / 2070** v1 runs complete, plus **v2 I1-I3** (aligned extraction). **Species:** H₂⁺ (2 u), H₂O⁺ (18 u), N₂⁺ (28 u) — the three ToF peaks in the paper. **Statistics:** 100000 secondaries per run; quoted σ is the RMS of **detected** x.

This report summarises the ion-mode parameter scan parallel to the e-mode replan (REPORT.md §8) and the fine C/D e-mode scan (§9). Combined e-mode + ion-mode write-up (includes v2 §4.5): CSNS_IPM_REPORT.md / CSNS_IPM_REPORT.pdf. Literature comparison (2006–2026): LITERATURE_REVIEW.md / LITERATURE_REVIEW.pdf. V2 plan and timing contract: SCAN_MATRIX_V2.md. Unlike e-mode, **no dense B-scan** was run: every block uses only $B_y \in 0, 200, 1000$ G (no guide / mid checkpoint / design 0.1 T), because closed ion scans already showed expansion **flat** vs B.

1. Scan matrix

Block	What is scanned	Beams	$\sigma_x=\sigma_y$	B [G]	Power [kW]
A	Reference beams	inj. 80 MeV; ext. 1.6 GeV; inj. 80 MeV	25 / 10 / 10 mm	0, 200, 1000	100–500
B	Beam size	inj.; ext.	3–20 mm (1 mm)	0, 200, 1000	100–500
C	Cage voltage	inj. only	10 mm	0, 200, 1000	100, 500
D	Beam offset	inj.; ext.	10 mm	0, 200, 1000	100, 500

Common settings: round beams ($\sigma_y=\sigma_x$); ideal cage $E_y = -V/231$ mm (ions collected at $y_{\min}$); circular 3-bunch train; simulation 12 μ s / 8000 Boris steps; $N_b = 7.8 \times 10^{12} \times (P/100 \text{ kW})$. SC-off reference: H₂⁺ @ 100 kW only (mass-independent without bunch fields, shared).
Re-run: ./scripts/run_imode_replan.sh · Evaluate: python3 scripts/evaluate_imode.py

2. Takeaways

- Guiding B does not fix ion-mode profiles.** Expansion is flat between 0 and 200 G and changes only slightly at 0.1 T — opposite to e-mode, where 200–300 G restores the profile.
- Ion distortion scales with power and inversely with beam size.** At 0.1 T, H₂⁺ injection at $\sigma=10$ mm goes from +82% (100 kW) to +422% (500 kW); at $\sigma=3$ mm, 500 kW gives obtained $\sigma \sim 56$ mm (+1700%+) for all three species.
- Lighter ions expand more at injection.** At 0.1 T / 100 kW / $\sigma=10$ mm: inj. H₂⁺ +82%, H₂O⁺ +64%, N₂⁺ +56%. At **aligned** extraction the 20 ns bunch is impulsive and H₂⁺ is the most distorted species (+104 % at 0 G, +89 % at 0.1 T); the v1 as-run extraction ordering (H₂⁺ +32 %, heavy ions +50–55 %) was a timing artefact.
- Cage voltage matters for ions (ToF / dwell time).** Expansion stays **positive** at every voltage — no e-mode-style sign flip. Higher V shortens drift and **reduces** the kick (H₂⁺ inj. 10 mm @ 1000 G: +198% at 5 kV → +70% at 30 kV).
- Beam offset:** Δx is translation-invariant; Δy (toward/away from the detector) changes expansion by a few percent.
- Extraction timing is aligned in v2 I1.** The v1 train arrived 125 ns late (generation at 80 ns, tracking at 204.6 ns). Re-running with tracking LongitudinalOffset = −80 ns gives H₂⁺ **+104.1 %**, H₂O⁺ +40.8 %, N₂⁺ +40.0 % at 10 mm / 100 kW / 0 G — matching the kick model (+102 / +40 / +39 %). Use those widths, not v1 as-run +33.5 %. Injection was already aligned. See CSNS_IPM_REPORT.md §4.5 and LITERATURE_REVIEW.md §3.5.

3. Block A — reference beams vs B

Three round reference beams at 0, 200, and 1000 G, all powers, all species.

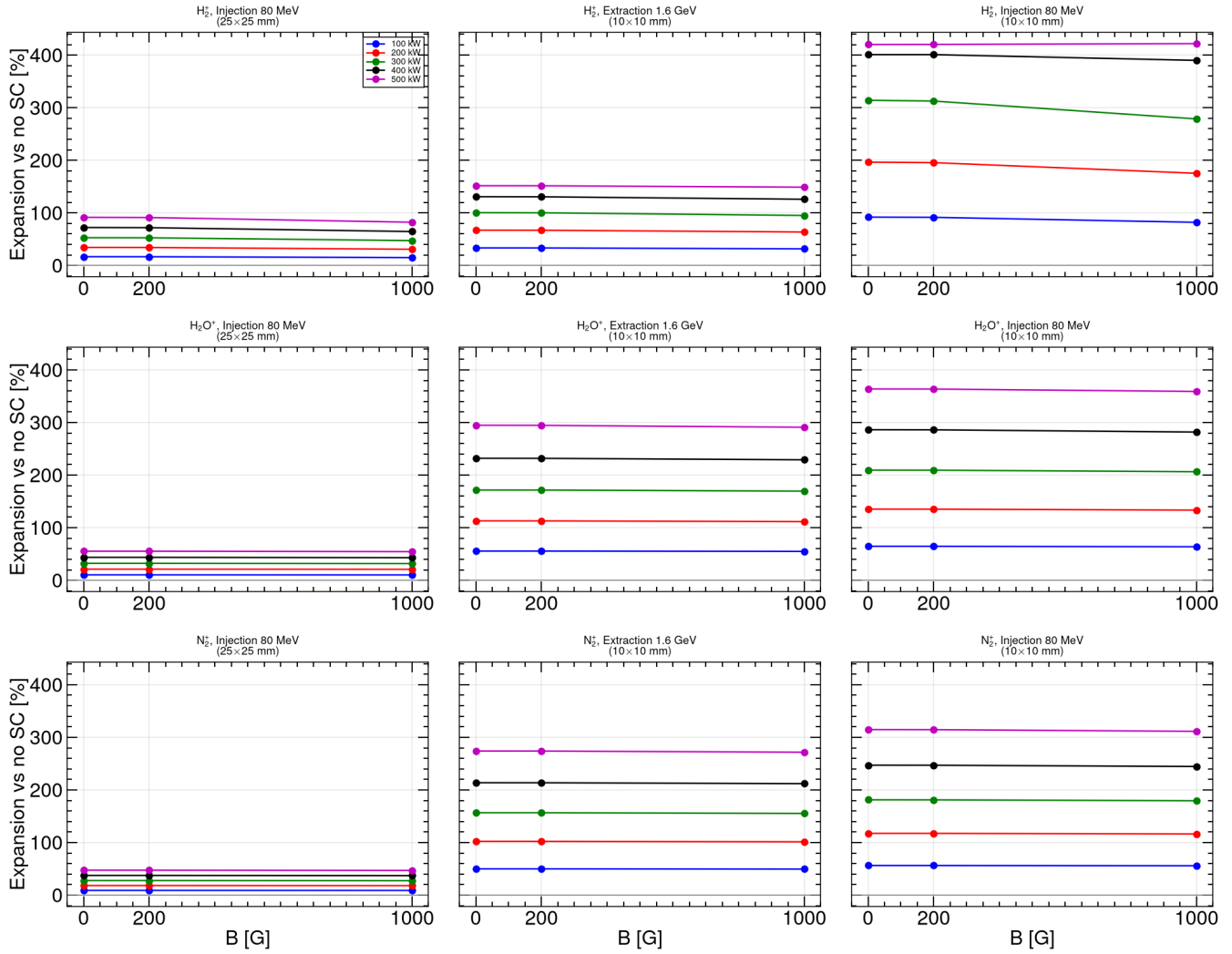


Figure 1. Profile expansion vs no-SC [%] for the three reference beams. Each panel is one species; columns are painted injection (25×25 mm), extraction (10×10 mm), and small injection (10×10 mm).

Expansion is **essentially independent of B** between 0 and 200 G (cyclotron radii are metres). The 0.1 T point is sometimes slightly lower but never restores the profile.

Table 1 — expansion at 100 kW [%]

Species	Inj. 25 mm 0 G	200 G	1000 G	Inj. 10 mm 0 G	200 G	1000 G	Ext. 10 mm 0 G†	200 G†	1000 G†
H ₂ ⁺	+16.7	+16.7	+15.0	+92.0	+91.6	+82.1	+33.5†	+33.4†	+31.6†
H ₂ O ⁺	+10.4	+10.4	+10.2	+64.7	+64.7	+63.8	+55.6†	+55.6†	+54.9†
N ₂ ⁺	+8.9	+8.9	+8.9	+56.3	+56.2	+55.7	+49.9†	+49.9†	+49.5†

† v1 as-run extraction (125 ns generation lag). **Superseded by v2 I1** (table below). Painted injection (25 mm) matches the elliptical-beam §1 numbers (+16% H₂⁺). Injection is correctly aligned in v1 (both centres at 480 ns).

Aligned extraction, 10 mm, 100 kW, 25 kV (v2 I1; tracking offset −80 ns):

Species	0 G aligned	0.1 T aligned	v1 as-run 0 G	Kick model aligned
H ₂ ⁺	+104.1 %	+89.1 %	+33.5 %	+102 %
H ₂ O ⁺	+40.8 %	+40.1 %	+55.6 %	+40 %
N ₂ ⁺	+40.0 %	+39.6 %	+49.9 %	+39 %

H_2^+ at extraction is the most distorted species once timing is correct. At 500 kW, 0 G, aligned 10 mm: H_2^+ +408 %, H_2O^+ +243 %, N_2^+ +235 %. Size / voltage / offset and the single-bunch check: CSNS_IPM_REPORT.md §4.5.

4. Block B — beam-size scan (3-20 mm)

Size scan at 0, 200, and 1000 G; injection and extraction; 100-500 kW.

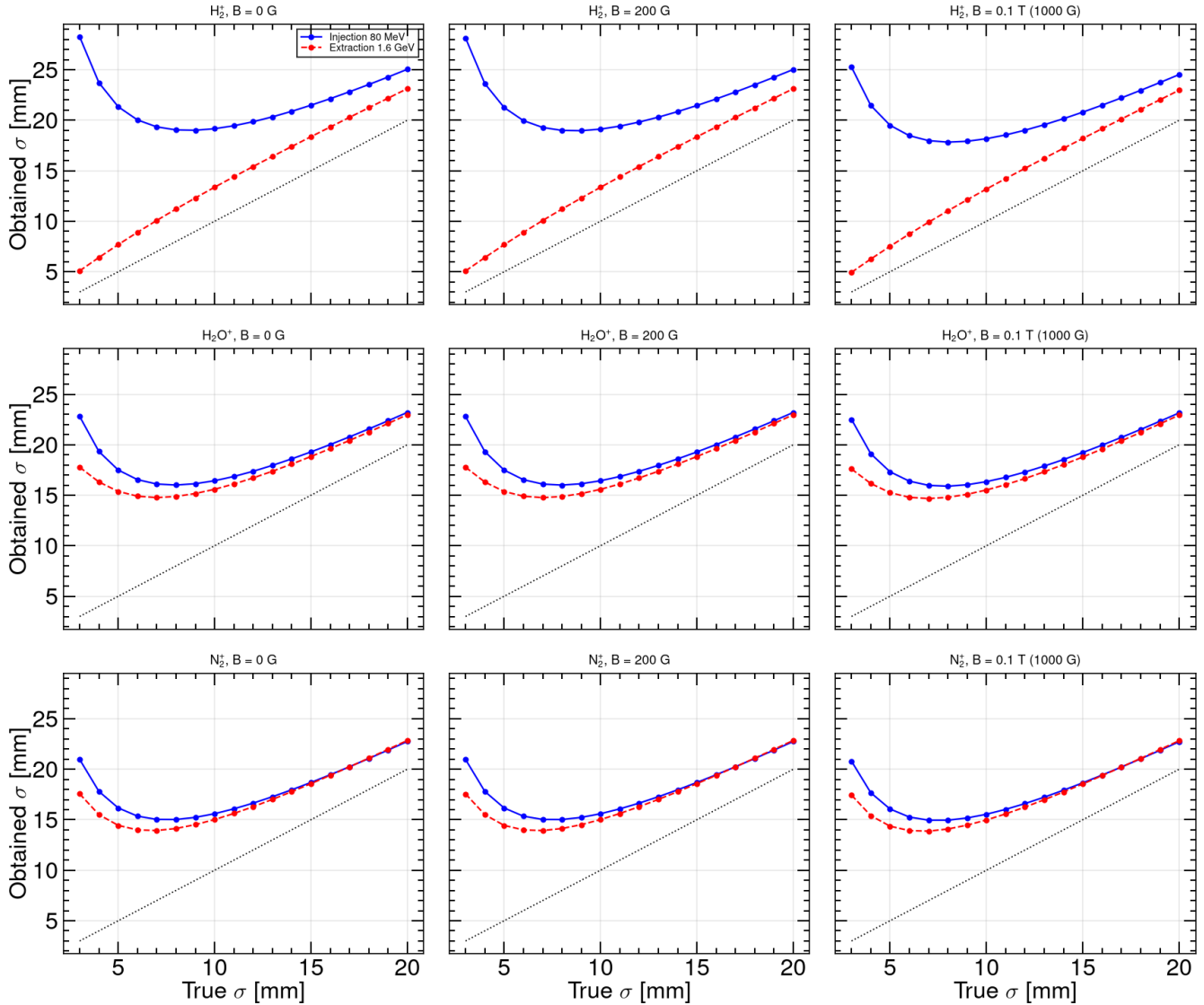


Figure 2. Obtained σ_x vs true σ_x at **0.1 T**, 100 kW. Dashed line: obtained = true. Every point lies **above** the diagonal at all three B fields — space charge always inflates the detected width.

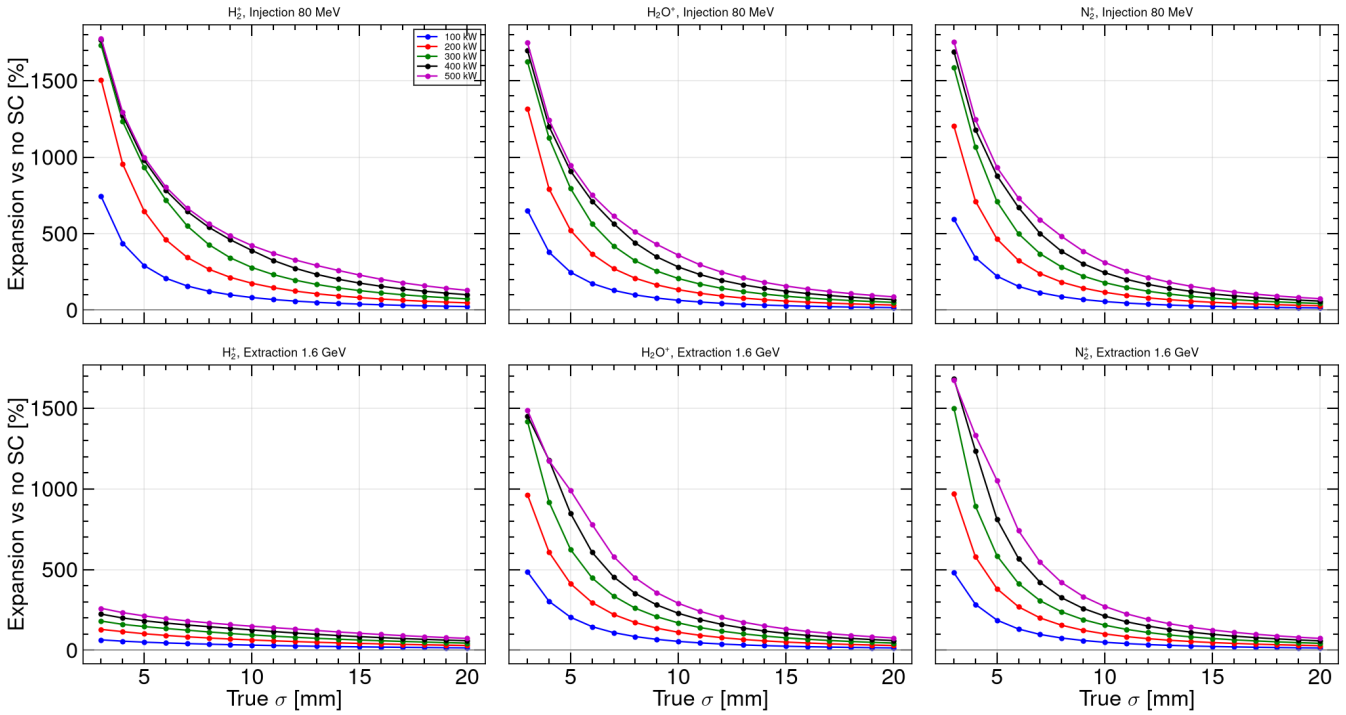


Figure 3. Expansion vs true σ_x at **0.1 T** for 100–500 kW. Smaller beams and higher power give dramatically larger expansion. On the v1 as-run extraction timing, injection looked worse than extraction for H_2^+ ; after the v2 alignment, extraction H_2^+ is larger than the v1 table (see §3).

Table 2 — obtained size at $\sigma_x = 10$ mm, 0.1 T, 100 kW

Species	Injection		Extraction†	
	obtained [mm]	expansion	obtained [mm]	expansion
H_2^+	18.2	+82%	13.2	+32%
H_2O^+	16.4	+64%	15.5	+55%
N_2^+	15.5	+56%	15.0	+50%

† v1 as-run. Aligned 0.1 T / 100 kW / 10 mm: H_2^+ 19.0 mm (+89 %), H_2O^+ / N_2^+ 14.0 mm (+40 %).

Table 3 — H_2^+ injection, $\sigma=10$ mm, 0.1 T vs power

100 kW	200 kW	300 kW	400 kW	500 kW
+82% (18.2 mm)	+175% (27.5 mm)	+279% (37.8 mm)	+390% (48.9 mm)	+422% (52.1 mm)

Worst case in the matrix: **injection, $\sigma=3$ mm, 500 kW, 0.1 T** — obtained $\sigma \approx 56$ mm (+1700%+) for all three species (cage-wall saturation).

5. Block C — cage voltage (5–30 kV)

Injection 10×10 mm; voltages 5, 10, 15, 20, 25, 30 kV; B = 0, 200, 1000 G; 100 and 500 kW.

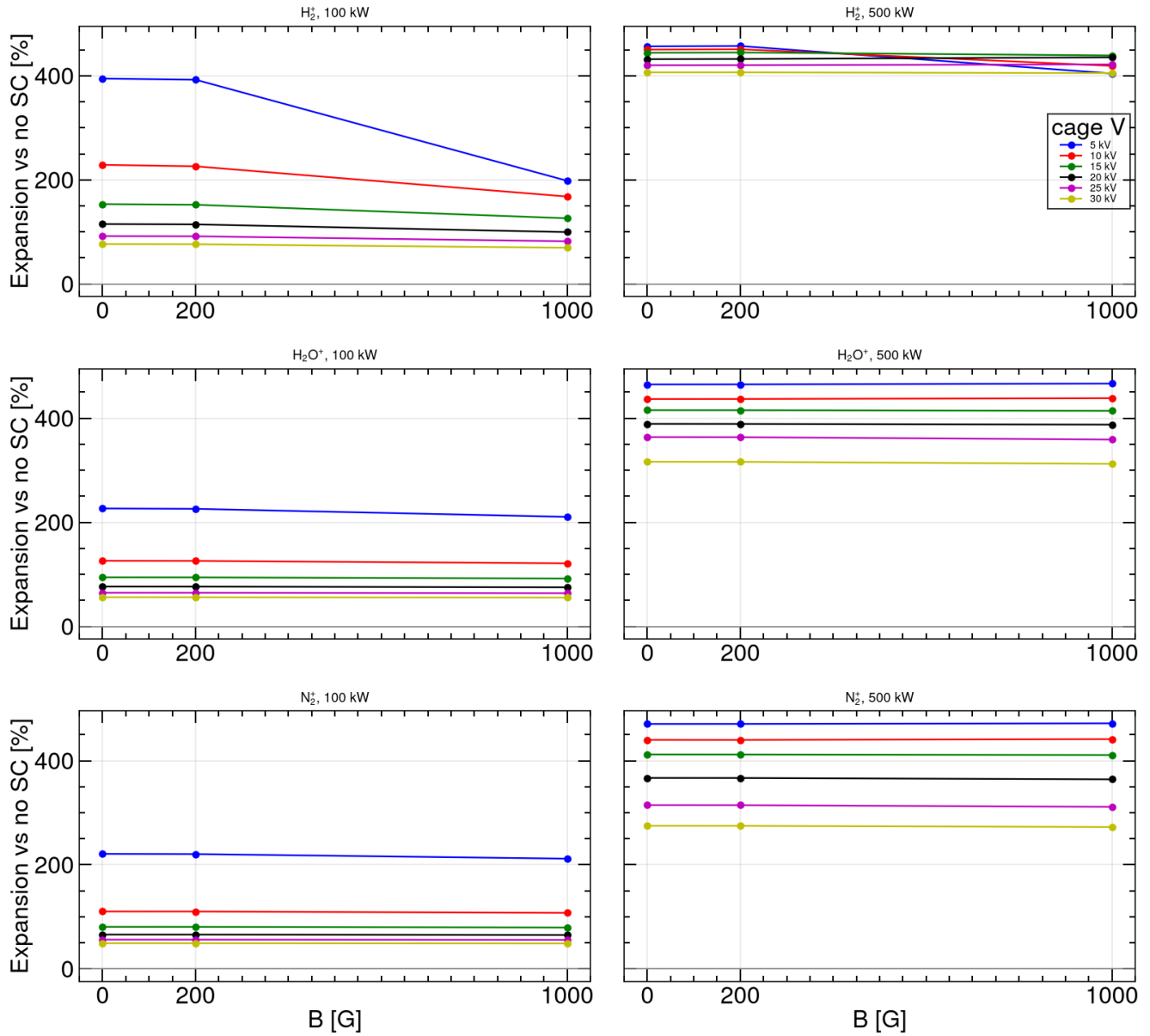


Figure 4. Expansion vs B for each cage voltage. Unlike e-mode Block C, ion expansion **never changes sign** — it stays positive at every V . Higher voltage shortens ion drift time and reduces the integrated space-charge kick.

Table 4 — H_2^+ injection 10 mm, 100 kW, 0.1 T vs cage voltage

5 kV	10 kV	15 kV	20 kV	25 kV	30 kV
+198%	+168%	+126%	+100%	+82%	+70%

At 500 kW the same trend holds but at much higher expansion (+300%–800% range).

6. Block D — beam offset

Injection and extraction 10×10 mm; offsets $(+5,0)$, $(+10,0)$, $(0,+5)$, $(0,-5)$ mm; $B = 0, 200, 1000$ G; 100 and 500 kW.

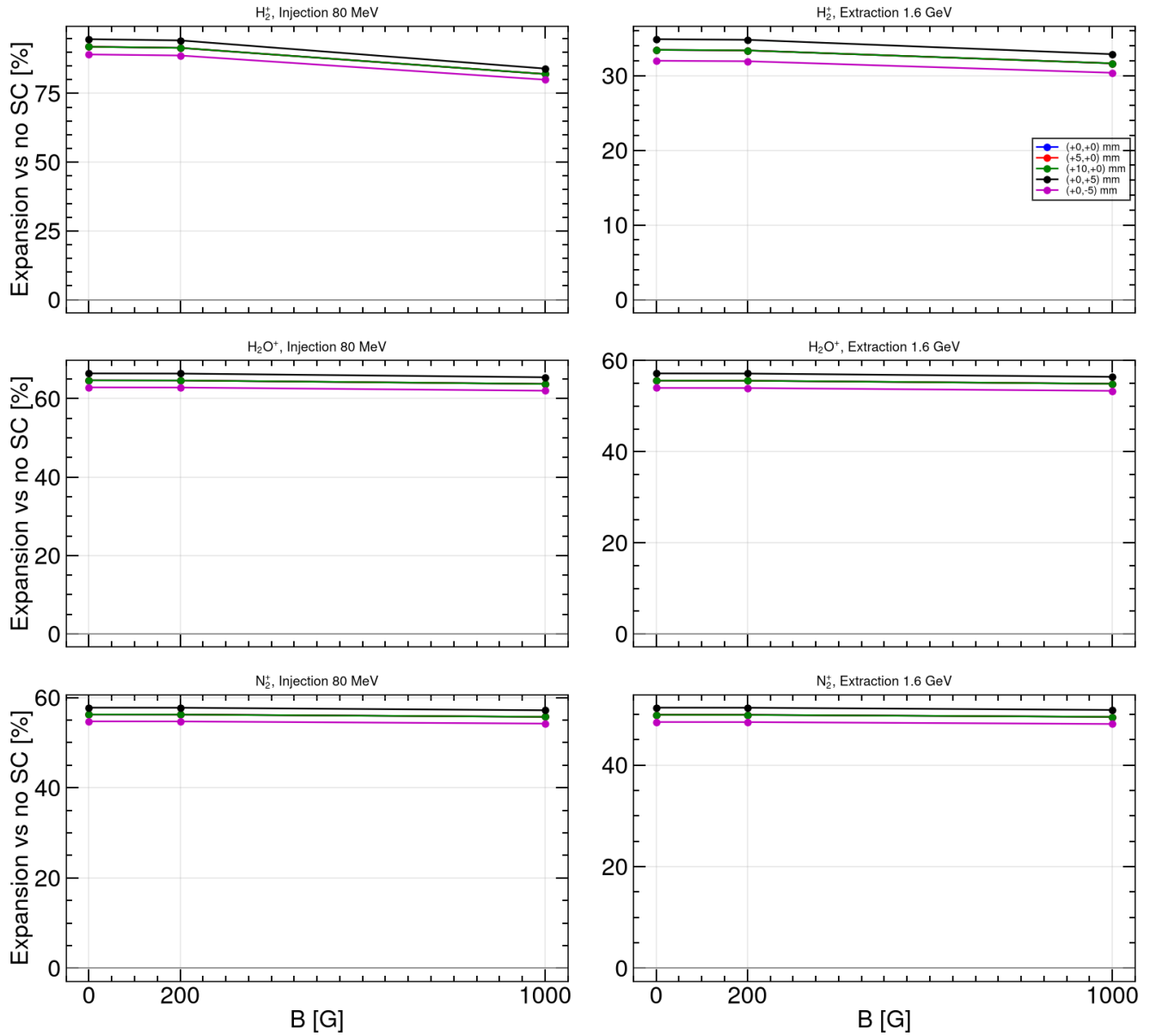


Figure 5. Expansion vs B for each offset (100 kW). $\Delta x = +5$ and $+10$ mm give **identical expansion** to the centred beam (translation invariance). $\Delta y = \pm 5$ mm changes expansion by a few percent (toward the detector: slightly less expansion).

Table 5 — H_2^+ injection 10 mm, 0.1 T, 100 kW vs offset

centred	$\Delta x = +5$	$\Delta x = +10$	$\Delta y = +5$	$\Delta y = -5$
+82.1%	+82.1%	+82.1%	+84.0%	+80.0%

Centroid shift vs no-SC is $\lesssim 0.6$ mm at extraction, 500 kW.

7. Comparison with e-mode (round-beam replan)

	E-mode (§8)	Ion-mode (this report)
Secondaries	electrons	H_2^+ / H_2O^+ / N_2^+
B grid	dense (5 G / 25 G)	0, 200, 1000 G only
Effect of B	oscillatory recovery; 300 G fixes profile	flat; no recovery
Cage voltage	sign flip at low B	always positive; V tunes magnitude
Beam offset	Δx invariant; Δy matters	same

Worst case	low B, wrong polarity	small σ , high P; aligned extraction H_2^+ or injection
Extraction ions	n/a	use v2 aligned widths (H_2^+ +104 % at 0 G / 10 mm / 100 kW)

Practical implication: tune the IPM guiding field and cage voltage for **faithful electron profiles**; treat **ion-mode size measurements as space-charge biased** unless a correction model is applied. Extraction ion widths must come from the aligned v2 table, not from v1 as-run +33.5 %.

8. Data products

File	Contents
`output/csns_imode_bscan_summary.csv`	Block A (135 rows)
`output/csns_imode_size_summary.csv`	Block B (1620 rows)
`output/csns_imode_voltage_summary.csv`	Block C (108 rows)
`output/csns_imode_offset_summary.csv`	Block D (180 rows)
`output/csns_v2_summary.csv`	v2 I1-I3 (aligned extraction, look-up, single bunch)
`plots/csns_imode_*.png`	Figures 1-5 above
`plots/csns_v2_aligned_extraction.png`	Aligned vs as-run vs kick model
`configs/csns_rcs_ipm/imode/README.md`	v1 matrix specification

CSV files under output/imode/ and output/v2/ (particle lists) are gitignored.